no “hair.” The flat spacetime exemplars include textbook calculations of how the mass of a black hole or other body changes when gravitational waves are captured by it, and calculations by Clifford Will, Thibault Damour, and others of how neutron stars orbiting each other generate gravitational waves (waves of shrinkage-producing field).

Each piece of a paradigm—its laws, its pictures, and its exemplars—is crucial to my own mental processes when I’m doing research. The pictures (mental and verbal as well as on paper) act as a general compass. They give me intuition as to how the Universe probably behaves; I manipulate them, along with mathematical doodlings, in search of interesting new insights. If I find, from the pictures and doodlings, an insight worth pursuing (for example, the hoop conjecture in Chapter 7), I then try to verify or refute it by careful mathematical calculations based on the paradigm’s mathematically formulated laws of physics. I pattern my careful calculations after the paradigm’s exemplars. They tell me what level of calculational precision is likely to be needed for reliable results. (If the precision is too poor, the results may be wrong; if the precision is too high, the calculations may eat up valuable time unnecessarily.) The exemplars also tell me what kinds of mathematical manipulations are likely to get me through the morass of mathematical symbols to my goal. Pictures also guide the calculations; they help me find shortcuts and avoid blind alleys. If the calculations verify or at least make plausible my new insight, I then communicate the insight to relativity experts by a mixture of pictures and calculations, and I communicate to others, such as readers of this book, solely with pictures—verbal pictures and drawings.

The flat spacetime paradigm’s laws of physics can be derived, mathematically, from the curved spacetime paradigm’s laws, and conversely. This means that the two sets of laws are different *mathematical representations* of the same physical phenomena, in somewhat the same sense as 0.001 and $1/1000$ are different mathematical representations of the same number. However, the mathematical formulas for the laws look very different in the two representations, and the pictures and exemplars that accompany the two sets of laws look very different.

As an example, in the curved spacetime paradigm, the verbal picture of Einstein’s field equation is the statement that “mass generates the curvature of spacetime.” When translated into the language of the flat spacetime paradigm, this field equation is described by the verbal picture “mass